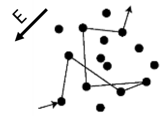
Detour: Drude model

Calculating R resistance from σ conductivity (L and A are the length and cross-section of the sample): $R = G^{-1} = \frac{1}{\sigma}\frac{L}{A}$

Ohm's law for current density: $\mathbf{j} = \sigma\mathbf{E}$

A simple microscopic model of electrical conduction (Drude model):



In a zero electric field, the average momentum of electrons is zero. In finite field, electrons gain momentum in the direction of the field strength (drift momentum). However, as they move through the crystal lattice, they undergo collisions on average at intervals τ, after which their momentum points in random directions (e.g. scattering by lattice defects, impurity atoms, lattice vibrations, etc.), i.e. the average drift momentum gained in the electric field between two collisions is lost by the electrons in time τ due to the collisions:

Average drift velocity gained between two collisions

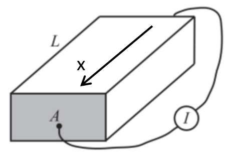
$$\mathbf{p}_{\text{drift}} = m\,\overbrace{\mathbf{v}_{\text{drift}}} = -e\mathbf{E}\tau \;\Rightarrow\; \mathbf{v}_{\text{drift}} = -\frac{e\tau}{m}\mathbf{E}$$

Key parameter: τ. The more collisions occur per unit time (i.e. the smaller τ), the greater the resistance.

$$\mathbf{j} = -ne\mathbf{v}_{\text{drift}} = \frac{ne^2\tau}{m}\mathbf{E} \;\Rightarrow \sigma = \frac{ne^2\tau}{m} \longleftarrow$$

conductivity in the Drude picture

The basic concept of thermal noise



In thermal equilibrium, without an external field, the average momentum of electrons is zero, but due to interaction with a finite temperature crystal lattice, electrons can gain random momentum, which is lost in time τ due to collisions. Thus, at a "snapshot", the total momentum of electrons moving in a given direction may be greater than that of electrons moving in the opposite direction. The momentum of the individual electrons on a snapshot is preserved on the time-scale of τ and lost afterwards. For this reason, in the arrangement shown in the figure (the two ends of a metal sample of length L with cross-section A are short-circuited), we will observe a random current fluctuation with a time average of zero.

Again, τ is the key parameter, the correlations in the current are lost on the characteristic time-scale of τ.

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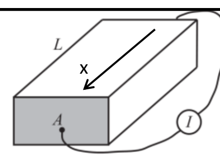
Calculation of thermal noise



Current density of a single electron in the x direction: $j_x(t) = -nev_x(t) = -\frac{1}{LA}ev_x(t);$

Current of a single electron: $I_1(t) = Aj_x(t) = -\frac{ev_x(t)}{L}$

The current-current correlation function for a single electron:

$$C_1(\Delta t) = \langle I_1(t)I_1(t+\Delta t)\rangle = \frac{e^2}{L^2}\langle v_x(t)v_x(t+\Delta t)\rangle \approx \frac{e^2}{L^2}\langle v_x^2(0)\rangle\exp(-\Delta t/\tau)$$

The collisions cause the correlation to be lost in characteristic time τ

The correlation function for N electrons, assuming that the electrons are independent of each other:

$$C(\Delta t) = NC_1(\Delta t) = \frac{Ne^2}{L^2}\langle v_x^2(0)\rangle\exp(-\Delta t/\tau) = \overbrace{\underbrace{\underbrace{\frac{N}{AL}}_{n}\frac{e^2\tau}{m}}_{\sigma}\frac{A}{L}}^{G}\frac{1}{\tau}\underbrace{m\langle v_x^2(0)\rangle}_{kT}\exp(-\Delta t/\tau) = G\frac{kT}{\tau}\exp(-\Delta t/\tau)$$

Equipartition theorem (will come from thermodynamics)

Previously we calculated the noise density for an exponentially decaying correlation function:

$$S_I(\omega) = \frac{4kTG}{1+\omega^2\tau^2} \underset{\omega<<1/\tau}{\approx} 4kTG$$

Later, we will see from solid state physics that we have miscalculated at two points:

$$m\langle v_x^2(0)\rangle \neq kT, \text{rather: } m\langle v_x^2(0)\rangle \approx \frac{mv_F^2}{3} \qquad \frac{mv_F^2}{2} = E_F$$

Fermi velocity *Fermi energy*

In addition, only about $N*kT/E_F$ of the total N electrons can be considered independent, the rest are "frozen" and contribute zero to the current. These two errors roughly compensate each other.

Despite the miscalculation, we have learned an important conclusion: the correlation time of noise is the same τ time between the collisions, which is the key parameter of the G conductance as well. So, it is understandable that thermal noise depends not only on temperature but also on conductance.

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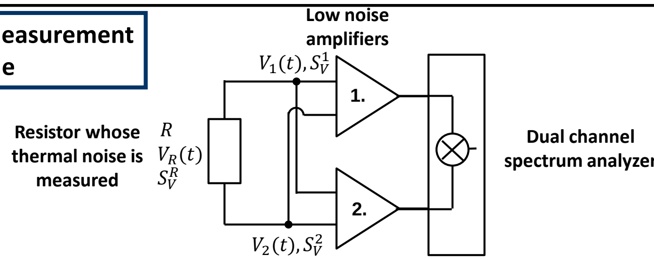
Cross-correlation measurement technique



The resistor's voltage fluctuates over time due to thermal noise: $V_R(t)$

The input noise of the amplifiers ($V_1(t)$ és $V_2(t)$) is added to the thermal noise of the resistor. The input noise of the two amplifiers is independent of each other! Like this, the two amplifiers output $A(V_R(t)+V_1(t))$ and $A(V_R(t)+V_2(t))$ voltage, respectively.
The dual-channel spectrum analyzer calculates a cross-correlation, i.e. it examines the spectrum of the product of two signals! As an illustration of the the method, let's just look at the expected value of the cross-product:

$$\langle A(V_R(t)+V_1(t))\cdot A(V_R(t)+V_2(t))\rangle = A^2\underbrace{\langle V_R^2(t)\rangle}_{S_V^R\cdot BW} + \underbrace{A^2\langle V_R(t)\rangle\langle V_2(t)\rangle + A^2\langle V_1(t)\rangle\langle V_R(t)\rangle + A^2\langle V_1(t)\rangle\langle V_2(t)\rangle}_{0}$$

If only one amplifier were used, the output voltage would have the mean squared deviation:

$$\langle A^2(V_R(t)+V_1(t))^2\rangle = A^2\underbrace{\langle V_R^2(t)\rangle}_{S_V^R\cdot BW} + A^2\underbrace{\langle V_1^2(t)\rangle}_{S_V^1\cdot BW}$$

That is, for a single amplifier, the sum of the noise density of the sample and the input noise density of the amplifier is measured, while for a cross-correlation of two amplifiers, the noise of the amplifiers is cancelled out. When measuring noise, the amplifier input noise is usually comparable to or greater than the noise under test.

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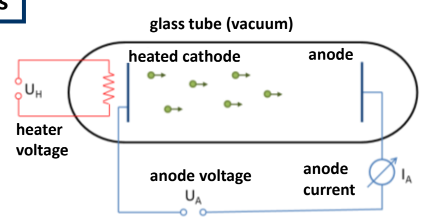
Shot noise of independently emitted electrons

The shot noise is related to the quantized electron charge, which results in current of individual "electron shots" instead of a constant noiseless current. Schottky made the first shot noise measurement in 1918 measuring the current noise of a vacuum tube. Electrons emitted randomly from a heated cathode are driven by the applied voltage to the anode, where a current is detected. In this case the electrons are emitted randomly and independently of each other, so the electron incidences are well described by the Poisson distribution:



$$\boxed{P_N(t) = \frac{t^N}{\tau^N N!}e^{-t/\tau}}$$

← The probability that N electrons are emitted within time t, such that the average time between the independent electron emissions is τ

[This formula can be proved by induction

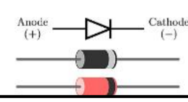
$$P_N(t+\mathrm{d}t) = P_{N-1}(t)\frac{\mathrm{d}t}{\tau} + P_N(t)\left(1-\frac{\mathrm{d}t}{\tau}\right) \Rightarrow \frac{\mathrm{d}P_N(t)}{\mathrm{d}t} = \frac{P_{N-1}(t)-P_N(t)}{\tau}, \ldots.$$]

A special property of the Poisson distribution: the mean value of N for a certain time is the same as its mean squared deviation:

$$\langle N\rangle = \sum N\cdot P_N(t) = \frac{t}{\tau}, \qquad \langle(\Delta N)^2\rangle = \langle N^2\rangle - \langle N\rangle^2 = \langle N\rangle$$

Note, that $\langle(\Delta N)^2\rangle$ is proportional to the current noise, whereas $\langle N\rangle$ is proportional to the mean current, i.e. the $S_I \sim I$ relation is expected for the shot noise of independently emitted electrons ("Poisson noise")

Independent electron emissions are not only available in a vacuum tube, but also in other systems, like a semiconductor diode at small enough currents.



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